

Modular Supervisory Control of Discrete Event Systems: Normality Under Masks

Shaopeng Hu ^a, Zhengkang Zhang ^a, Jan Komenda ^{b,*}, Zhiwu Li ^{a,c}

^a*School of Electro-Mechanical Engineering, Xidian University, Xi'an 710071, China*

^b*Institute of Mathematics of the Czech Academy of Sciences, 115 67 Prague, Czechia*

^c*Institute of Systems Engineering, Macau University of Science and Technology, Taipa, Macau*

Abstract

This paper investigates modular supervisory control of discrete event systems (DESs) under mask-based partial observation. While the supervisory control of modular DESs with partial observation has been studied under natural projections, it has not yet been addressed under mask functions, where two or more observable events may be indistinguishable and represented by the same symbol in a new alphabet of observable labels. Such general observations naturally arise when different types of events cannot be distinguished, due to limited sensor capabilities, or as a result of cyber-attacks that modify observable events. We propose sufficient conditions to guarantee maximal permissiveness of local control synthesis. In particular, we show that the commutativity between local projections and masks, together with the injectivity of relabeling on shared events and the observability of all shared events, forms such sufficient conditions. Furthermore, this result is extended to global (non-decomposable) specifications. An illustrative example is presented to demonstrate the proposed approach.

Key words: Discrete event system, modular supervisory control, normality, mask-based partial observation.

1 Introduction

Discrete event systems (DESs) [Cassandras and Lafortune, 2021] play a fundamental role in modeling and analyzing modern technological systems, particularly in engineering domains such as flexible manufacturing and transportation. These systems are often composed of multiple interacting components, which limits the applicability of standard monolithic supervisory control architectures. Moreover, the inherent computational complexity [Ramadge and Wonham, 1987] motivates the use

of modular and distributed control architectures [Wonham and Cai, 2018].

Modular supervisory control of DESs was first introduced by Wonham and Ramadge [1988], where modularity refers to handling multiple specifications whose intersection defines the overall specification. The objective of modular supervisory control is to synthesize a non-blocking and maximally permissive supervisor based on local plants, without explicitly constructing the global (monolithic) system. Two major challenges in modular supervisory control are conflicts among local supervisors and the potential loss of maximal permissiveness, which is sometimes traded off for reduced computational complexity in the modular synthesis.

Modular supervisory control has been extensively studied over the past decades. Willner and Heymann [1991] propose a sufficient condition that all shared events are controllable to guarantee that the maximally permissive local supervisors working in parallel achieve the same controlled language as the monolithic maximally

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* Corresponding author.

Email addresses: sphu@stu.xidian.edu.cn (Shaopeng Hu), zhengkang@stu.xidian.edu.cn (Zhenkang Zhang), komenda@ipm.cz (Jan Komenda), zwli@must.edu.mo (Zhiwu Li).

permissive supervisor. Building on this, Flordal et al. [2007] develop an incremental compositional approach that avoids constructing the monolithic supervisor while preserving non-blockingness and maximal permissiveness of the closed-loop system. Nevertheless, both approaches assume full observability of the plant and a decomposable specification.

In the context of partial observation, unobservable events significantly increase the complexity of supervisory control, making it more challenging to guarantee maximal permissiveness of the combined actions of local supervisors. When all controllable events are observable, maximally permissive supervisors can be obtained as the supremal controllable and normal sublanguages of the specification. A key question then is under which conditions a modularly synthesized supervisor coincides with the monolithic (global) supervisor. In DESs with partial observation, global (non-decomposable) specifications have been studied in Komenda et al. [2012], and modular control is addressed in Komenda and Masopust [2024]. Specifically, Komenda and Masopust [2024] demonstrate that, under the modified observation consistency property, a global maximally permissive normal supervisor can be realized locally as the parallel composition of local normal supervisors. However, verifying this property is PSPACE-hard. Therefore, an easily verifiable sufficient condition, namely that all shared events are observable, has been proposed to guarantee modified observation consistency Komenda and Masopust [2023].

In this paper, we investigate modular supervisory control under mask-based observations as introduced in Kumar and Garg [1995], Kumar et al. [2005], Cho and Marcus [1989]. Unlike the widely used natural projection, a mask allows multiple observable events to be mapped to a single label from a possibly different observation alphabet. In other words, a mask can not only remove unobservable events but can also rewrite observable events, introducing additional indistinguishability challenges. To the best of our knowledge, modular supervisory control under mask-based observations has received little attention. To this end, we extend the framework of modular supervisory control from natural projection-based observations to the more general setting of mask-based observations.

There are several motivations to extend modular supervisory control from projection-based to mask-based observations. First, in distributed supervisory synthesis, it is natural to handle both event erasure and event relabeling. Second, in the context of cyber-security, masks can naturally model sensor attacks, where the events observable by the supervisor may be deleted or replaced by an attacker. In such scenarios, observation masks should be incorporated both in the system and in the attack model to ensure attack stealthiness. The main contributions of this paper are summarized as follows.

- We establish a necessary and sufficient condition for the commutativity between observation masks and local projections in modular DESs based on factorization of masks in terms of relabeling and natural projections.
- For modular DESs with a decomposable specification, we derive sufficient conditions to compute the global supremal normal sublanguages w.r.t. a plant and its mask without constructing the global plant.
- For more general settings involving global (non-decomposable) specifications, we provide sufficient conditions to guarantee the modular synthesis of supremal controllable and normal sublanguages under mask-based observations.

The organization of the paper follows. The preliminaries with basic concepts of deterministic finite automata and modular supervisory control are in Section II. Section III introduces modular supervisory control under masks and presents the modular supervisor synthesis method for the supremal normal sublanguages. In Section IV, the results are extended to global specifications and to the issues of both controllability and normality. Finally, Section V concludes this research.

2 Preliminaries

This section overviews the related definitions and results from supervisory control of deterministic finite automata [Cassandras and Lafortune, 2021, Wonham and Cai, 2018] and the recent results on modular supervisory theory for the subsequent development [Komenda and Masopust, 2024].

2.1 Strings, languages, and automata

The cardinality of a set A is denoted by $|A|$. An alphabet, Σ , is a finite nonempty set of events. The set of finite strings over Σ is denoted by Σ^* , including the empty string denoted by ε . The length of a string $s \in \Sigma^*$ is symbolized by $|s|$. The set of prefixes of $s \in \Sigma^*$ is defined as $\bar{s} = \{s' \in \Sigma^* \mid \exists t \in \Sigma^* : s = s't\}$. A language is a subset of Σ^* . The set of prefixes of a language L is denoted by $\bar{L} = \bigcup_{s \in L} \bar{s}$. A language L is *prefix-closed* if $L = \bar{L}$.

A *projection* $P : \Sigma^* \rightarrow \Sigma'^*$ with $\Sigma' \subseteq \Sigma$, is defined as follows: $P(\varepsilon) = \varepsilon$; $P(\sigma) = \varepsilon$ if $a \in \Sigma \setminus \Sigma'$; $P(\sigma) = \sigma$ if $\sigma \in \Sigma'$; $P(s\sigma) = P(s)P(\sigma)$ for $s \in \Sigma^*$ and $\sigma \in \Sigma$. The action of P on a string $\sigma_1\sigma_2 \cdots \sigma_n$ is to remove the events that are not in Σ' . The inverse projection of a string $w \in \Sigma'^*$ under the projection P is the set $P^{-1}(w) = \{s \in \Sigma^* \mid P(s) = w\}$. Both P and P^{-1} can be extended to languages in the standard way based on the set union.

A *deterministic finite automaton* (DFA) is a quintuple $G = (Q, \Sigma, \delta, q_0, Q_m)$, where Q is a finite set of states, Σ

is a finite set of events, $q_0 \in Q$ is an initial state, $Q_m \subseteq Q$ is the subset of marker states, and $\delta: Q \times \Sigma \rightarrow Q$ is a (partial) transition function. As usual, δ can be extended to $\delta: Q \times \Sigma^* \rightarrow Q$, where Σ^* is the set of all finite-length strings, including the empty string ε . We write $G = (Q, \Sigma, \delta, q_0)$ if the set of marked states Q_m is of no interest. The language and marked language of G are defined by $L(G) = \{s \in \Sigma^* \mid \delta(q_0, s) \in Q\}$ and $L_m(G) = \{s \in L(G) \mid \delta(q_0, s) \in Q_m\}$, respectively.

For a DFA G , and another alphabet (of mask labels) Δ , a mask $M: \Sigma \rightarrow \Delta \cup \{\varepsilon\}$ is a function that either relabels an event $\sigma \in \Sigma$ into a symbol in $M(\sigma) \in \Delta$ or deletes it, resulting the empty string $M(\sigma) = \varepsilon$. Mask M can be extended to strings as usual, defined as $M: \Sigma^* \rightarrow \Delta^*$, that is, $M(se) = M(s)M(e)$ and $M(\varepsilon) = \varepsilon$, where $e \in \Sigma$ and $s \in \Sigma^*$. The inverse image of a string $w \in \Delta^*$ under the mask M is defined as $M^{-1}(w) = \{s \in \Sigma^* \mid M(s) = w\}$. The definitions of M and M^{-1} can be extended to languages as usual.

2.2 Basic concepts of supervisory control under masks

Given a DES G and a mask M , the set of events Σ is partitioned into two disjoint subsets, i.e., $\Sigma = \Sigma_o \cup \Sigma_{uo}$, where $\Sigma_o = \{\sigma \in \Sigma \mid M(\sigma) \in \Delta\}$ is the set of observable events and its complement, i.e., $\Sigma_{uo} = \{\sigma \in \Sigma \mid M(\sigma) = \varepsilon\}$, is that of unobservable events. From the perspective of controllability, Σ can be divided into the set of controllable events Σ_c and that of uncontrollable events Σ_{uc} , i.e., $\Sigma = \Sigma_c \cup \Sigma_{uc}$. A language $K \subseteq L(G)$ is *controllable* w.r.t. $L(G)$ and the set Σ_{uc} of uncontrollable events if $\bar{K}\Sigma_{uc} \cap L(G) \subseteq \bar{K}$. Language K is *observable* w.r.t. $L(G)$, M , and Σ_c if, for every $s \in \bar{K}$ and every $\sigma \in \Sigma_c$, $s\sigma \notin \bar{K}$ and $s\sigma \in L(G)$ imply $M^{-1}[M(s)]\sigma \cap \bar{K} = \emptyset$. Language K is *normal* w.r.t. $L(G)$ and the mask $M: \Sigma^* \rightarrow \Delta^*$ if for all $s \in \bar{K}$ and $s' \in L(G)$ such that $M(s') = M(s)$, $s' \in \bar{K}$ holds, equivalently $M^{-1}M(\bar{K}) \cap L = \bar{K}$.

The supremal normal sublanguage of K w.r.t. $L(G)$ and the mask M , denoted by $\text{supN}(K, L(G), M)$, is equal to the union of all sublanguages of K that are normal w.r.t. $L(G)$ and M . Similarly, we denote by $\text{supCN}(K, L(G), \Sigma_{uc}, M)$ the supremal controllable and normal sublanguage of K w.r.t. $L(G)$, Σ_{uc} , and M .

Given a DES G over Σ and a mask $M: \Sigma^* \rightarrow \Delta^*$, we denote the set of control patterns by $\Gamma = \{\gamma \subseteq \Sigma \mid \Sigma_{uc} \subseteq \gamma\}$. A *supervisor* of G w.r.t. a set of control patterns Γ is a map $S: M(L(G)) \rightarrow \Gamma$ returning, for every observed string of labels, a set of enabled events that always include all uncontrollable events. The *closed-loop system* of G under the supervision of S is the minimal language $L(S/G)$ such that $\varepsilon \in L(S/G)$ and $sa \in L(S/G)$ iff $s \in L(S/G)$, $sa \in L(G)$, and $a \in S(M(s))$. If the closed-loop system is non-blocking, i.e., $L_m(S/G) = L(S/G)$,

then S is said to be non-blocking. Intuitively, a supervisor disables certain controllable transitions according to the mask-based observation of the system.

2.3 Modular supervisory control

Many engineering systems are modeled as a composition of several subsystems, for DES typically as a synchronous product. The synchronous product of languages L_i over Σ_i is the language $\|_{i=1}^n L_i = \bigcap_{i=1}^n P_i^{-1}(L_i)$ over $\Sigma = \bigcup_{i=1}^n \Sigma_i$, where $P_i: \Sigma^* \rightarrow \Sigma_i^*$ is the projection to local alphabet Σ_i for $i = 1, \dots, n$. For DFAs G_i over Σ_i , there is a DFA $G = \|_{i=1}^n G_i$ over Σ satisfying $L(\|_{i=1}^n G_i) = \|_{i=1}^n L(G_i)$ and $L_m(\|_{i=1}^n G_i) = \|_{i=1}^n L_m(G_i)$. The languages L_i are *nonconflicting* if $\|_{i=1}^n L_i = \|_{i=1}^n \bar{L}_i$.

Given DFAs $G_1, \dots, G_n$ generating languages $L_i = L(G_i)$ with the global behavior $L = \|_{i=1}^n L_i$, and a specification $K \subseteq L$, the objective of the modular control problem is to synthesize local supervisors S_i such that $\|_{i=1}^n L(S_i/G_i) = L(S/\|_{i=1}^n G_i)$, where S is a maximally permissive supervisor for the specification K and for the global plant language L .

Let $P_i: \Sigma^* \rightarrow \Sigma_i^*$ denote the projections to modules, and let the corresponding observations and projections of modules be $P_{i,o}^i: \Sigma_i^* \rightarrow (\Sigma_i \cap \Sigma_o)^*$ and $P_{i,o}^o: \Sigma_o^* \rightarrow (\Sigma_i \cap \Sigma_o)^*$, respectively, as shown in Fig. 1. The locally observable events are represented by $\Sigma_{i,o} = \Sigma_i \cap \Sigma_o$. Given a modular DES $G = \|_{i=1}^n G_i$, with G_i over Σ_i , the set of shared events is defined by $\Sigma_s = \bigcup_{i \neq j} (\Sigma_i \cap \Sigma_j)$.

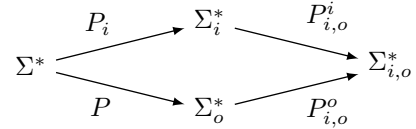


Fig. 1. Projection notations.

The following concept plays a key role in modular control with natural projection-based observation.

Definition 1 [Komenda and Masopust, 2024] A prefix-closed language $L \subseteq \Sigma^*$ is *modified observation consistent* w.r.t. projections P_i , P , and $P_{i,o}^i$ if, for every string $s \in L$ and $t' \in P_i(L)$ with $P_{i,o}^i(P_i(s)) = P_{i,o}^i(t')$, there exists a string $s' \in L$ such that $P(s) = P(s')$ and $P_i(s') = t'$. $\diamond$

Intuitively, by Definition 1, for every global string s and every local string t' that look locally the same as s , there exists a global string s' corresponding to t' in the local subsystem that looks the same as s .

3 Synthesis of normal supervisors for modular DESs under masks with local specifications

This section provides sufficient conditions to enforce modular normality under masks for local specifications directly from local models, avoiding thereby the explicit construction of the global plant.

3.1 Modular supervisory control under masks

Compared with natural projections, mask-based observations introduce additional challenges due to the potential renaming of events in a non-injective manner. For example, two strings that are observationally indistinguishable (mask-equivalent) over the global alphabet may not remain indistinguishable when projected onto local alphabets unless certain conditions are imposed. In contrast, natural projections always preserve indistinguishability under projection. This discrepancy arises because, for observation masks, commutativity between masks and local projections may not hold, whereas standard natural projections always commute with projections to modules.

Given a DFA $G = \parallel_{i=1}^n G_i$ over $\Sigma = \bigcup_{i=1}^n \Sigma_i$, let $M_i : \Sigma_i^* \rightarrow \Delta_i^*$ be the masks in modules, and let $P_i^\Delta : \Delta_i^* \rightarrow \Delta_i^*$ be the local projections of masks. Note that mask $M : \Sigma^* \rightarrow \Delta^*$ can be factorized as composition $M = R \circ P$ of natural projection $P : \Sigma^* \rightarrow \Sigma_o^*$ followed by the relabeling function $R : \Sigma_o^* \rightarrow \Delta^*$, defined by $R(\sigma) = M(\sigma)$ for all $\sigma \in \Sigma_o$. Similarly, the relabeling function of modules G_i can be introduced as $R_i : \Sigma_{i,o}^* \rightarrow \Delta_i^*$ with $\Delta_i = R_i(\Sigma_{i,o})$ such that $M_i = R_i \circ P_{i,o}^i$. We illustrate the relationships of these functions through the following diagram.

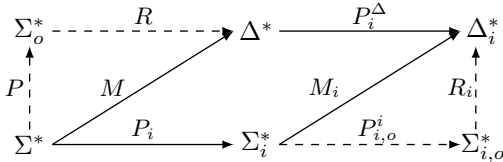


Fig. 2. Diagram of projections and masks.

Unlike the commutativity property illustrated in Fig. 1, which always holds for natural projections, the commutativity illustrated in Fig. 2 is not guaranteed for arbitrary choices of the mask and the relabeling function; rather, Fig. 2 depicts a particular mask M and relabeling function R for which the diagram commutes. Since several subsequent results rely on the commutativity in Fig. 2, it is essential that masks commute with local projections. In Lemma 5, we establish a necessary and sufficient condition for this commutativity, which essentially requires that the relabeling should preserve membership in the local alphabets.

Remark 2 Note that two events (or strings) sharing the same mask may be projected onto events (or strings) with different masks in some or all local alphabets. However, the commutativity illustrated by solid lines in Fig. 2 guarantees that for any two strings s, s' with $M(s) = M(s')$, we have $M_i(P_i(s)) = P_i^\Delta(M(s)) = P_i^\Delta(M(s')) = M_i(P_i(s'))$. This result will be formally proved in the next subsection (cf. Lemma 7). Moreover, every mask (global and local) can be factorized as the composition of a projection with a relabeling, i.e., $M = R \circ P$. $\diamond$

Based on the commutativity of the diagram in Fig. 2, the first problem to be addressed in this paper is stated.

Problem 3 Let $G = G_1 \parallel \dots \parallel G_n$ be a modular plant and $K = \parallel_{i=1}^n K_i$ a decomposable specification. Find sufficient conditions under which the global supremal normal sublanguage of K w.r.t. L and mask M equals the composition of local supremal normal sublanguages w.r.t. $L(G_i)$ and M_i , i.e., $\text{supN}(K, L, M) = \parallel_{i=1}^n \text{supN}(K_i, L_i, M_i)$. $\diamond$

Note that the supremal normal sublanguage of the specification here corresponds to the maximally permissive supervisor, representing the largest language that satisfies the supervisory control objectives under partial observation, assuming all controllable events are observable. To solve Problem 3, we adopt the following three assumptions to ensure $\text{supN}(K, L, M) = \parallel_{i=1}^n \text{supN}(K_i, L_i, M_i)$ holds.

- A1 For any $\sigma \in \Sigma_o$, $\sigma \in \Sigma_i$ iff $R(\sigma) \in \Delta_i$.
- A2 For all observed shared events $\sigma_1, \sigma_2 \in \Sigma_s \cap \Sigma_o$ such that $\sigma_1 \neq \sigma_2$, it holds $R(\sigma_1) \neq R(\sigma_2)$.
- A3 All shared events are observable, i.e., $\Sigma_s \subseteq \Sigma_o$.

Note that Assumptions A1 and A2 are additional properties of R to guarantee that modified observation consistency holds for mask-based observations.

Remark 4 Assumption A1 ensures commutativity between relabeling and local projections in Lemma 5, thereby preserving the local event membership. Moreover, Assumption A1 also requires that relabeling should not alter the local status of events, i.e., their membership in local alphabets. This is natural, since changing the distribution of events in the original Σ would affect synchronizations, causing the relabeled system to behave differently from the original system. Assumption A2 is used to prove half-distributivity of relabeling w.r.t. synchronous product in Lemma 8, providing an easily checkable condition required for maximal permissiveness under observation masks. Assumption A3 is a standard condition in supervisory control with natural projections, known to guarantee maximal permissiveness in that context. In addition, Assumption A2 can be viewed as the natural counterpart of Assumption

A3 when all events are observable, but some observable events are indistinguishable.

3.2 Solution procedure

This subsection presents an approach to the supervisor synthesis method for the supremal normal sublanguage of modular DESs under masks. We first show a sufficient condition for the commutativity of the masks and natural projections. As a first step, we consider the special case of observable strings, where (observation) projections P and $P_{i,o}^i$ are identity maps. Then $M = R$ and $M_i = R_i$, which means that the diagram collapses to $R_i \circ P_i(w) = P_i^\Delta R(w)$ for $w \in \Sigma_o^*$.

Lemma 5 Let $\Sigma = \cup_{i=1}^n \Sigma_i$ be a distributed alphabet, with M and M_i denoting the global and local masks, and R and R_i the corresponding relabeling functions, as in Fig. 2. Then $R_i P_i(w) = P_i^\Delta R(w)$ holds for all $w \in \Sigma_o^*$ iff Assumption A1 holds.

Proof: (If) We use structural induction on string w to show that $R_i P_i(w) = P_i^\Delta R(w)$ for all $w \in \Sigma_o^*$. The base step $w = \varepsilon$ is trivial as both sides equal ε . The induction hypothesis is that $R_i P_i(w) = P_i^\Delta R(w)$ for w and we will show that for $\sigma \in \Sigma_o$, we also have $R_i P_i(w\sigma) = P_i^\Delta R(w\sigma)$. Since all projections and relabeling maps are catenative, i.e., $R_i P_i(w\sigma) = R_i P_i(w) R_i P_i(\sigma)$ and $P_i^\Delta R(w\sigma) = P_i^\Delta R(w) P_i^\Delta R(\sigma)$ it suffices to show that $R_i P_i(\sigma) = P_i^\Delta R(\sigma)$. We distinguish the cases (i) $\sigma \in \Sigma_i$ and (ii) $\sigma \in \Sigma \setminus \Sigma_i$. In the first case, we have by Assumption A1 that $R_i(\sigma) \in \Delta_i$. Thus, $R_i P_i(\sigma) = R_i(\sigma) = P_i^\Delta R_i(\sigma) = P_i^\Delta R(\sigma)$. In the second case, we have by Assumption A1 that $R_i(\sigma) \notin \Delta_i$. It holds $R_i P_i(\sigma) = R_i(\varepsilon) = \varepsilon = P_i^\Delta R(\sigma)$.

(Only if) We first show that given $\sigma \in \Sigma_o \cap \Sigma_i$, $R(\sigma) \in \Delta_i$ holds. From the diagram of Fig. 2, given $\sigma \in \Sigma_{i,o}$, we have $R_i(\sigma) \in \Delta_i$, suggesting $R(\sigma) \in \Delta_i$. Now we prove that given $\sigma \in \Sigma_o$ such that $R(\sigma) \in \Delta_i$, $\sigma \in \Sigma_i$ holds. From the diagram of Fig. 2, $P_i^\Delta R(\sigma) = R_i P_i(\sigma) = R_i(\sigma)$. It holds $R_i(\sigma) \in \Delta_i$, which implies $\sigma \in \Sigma_{i,o}$ and thus $\sigma \in \Sigma_i$. $\square$

Remark 6 Lemma 5 establishes a necessary and sufficient condition for the commutativity between relabeling of observable events and natural projections. The full diagram in Fig. 2 commutes iff Assumption A1 holds. Indeed, in the induction step of the previous proof, for the case $\sigma \in \Sigma_{uo}$, it always holds $M_i P_i(\sigma) = R_i P_{i,o}^i P_i(\sigma) = R_i(\varepsilon) = \varepsilon = P_i^\Delta R P(\sigma) = P_i^\Delta M(\sigma)$.

Building on this, the following lemma shows that Assumption A1 ensures that global indistinguishability implies local indistinguishability.

Lemma 7 Let $G = \cup_{i=1}^n G_i$ be a modular DES over $\Sigma = \cup_{i=1}^n \Sigma_i$ and M be a mask. For all strings $s, s' \in L = L(G)$

with $M(s) = M(s')$, it holds $M_i(P_i(s)) = M_i(P_i(s'))$ if Assumption A1 holds.

Proof: For all strings $s, s' \in L$ with $M(s) = M(s')$, we have $M_i(P_i(s)) = P_i^\Delta M(s) = P_i^\Delta M(s') = M_i(P_i(s'))$, which completes the proof. $\square$

The following lemma presents a sufficient condition that guarantees maximal permissiveness of modular supervisors under mask-based observations, ensuring that local synthesis aligns with global specifications.

Lemma 8 If $\Sigma = \cup_{i=1}^n \Sigma_i$ is a distributed alphabet with the shared event set $\Sigma_s = \cup_{j \neq i} \Sigma_i \cap \Sigma_j$, then for the languages over observable alphabets $L_i \subseteq \Sigma_{i,o}^*$ under Assumptions A1 and A2, it holds $\|_{i=1}^n R_i(L_i) \subseteq R(\|_{i=1}^n L_i)$.

Proof: Let $s \in \|_{i=1}^n R(L_i)$, i.e., for all $i = 1, 2, \dots, n$ with $P_i^\Delta(s) \in R(L_i)$, there exist strings $w_i \in L_i$, $i = 1, \dots, n$ such that $P_i^\Delta(s) = R(w_i)$. We need to show that there exists $w \in \|_{i=1}^n L_i$ such that $s = R(w)$. It is natural to choose $w \in \|_{i=1}^n w_i$, since then one has $w \in \|_{i=1}^n L_i$. We first show that $\|_{i=1}^n w_i \neq \emptyset$. By $s \in \|_{i=1}^n P_i^\Delta(s) \neq \emptyset$, we know that $\|_{i=1}^n R_i(w_i) \neq \emptyset$. We will show that due to Assumption A2, it implies $\|_{i=1}^n w_i \neq \emptyset$ as well. Note that both compositions are nonempty if and only if the projections to their respective shared events are. We use the following notation. Let $\Delta_s = \cup_{j \neq i} \Delta_i \cap \Delta_j$ be the set of shared mask labels, and let the corresponding natural projections be $P_s : \Sigma^* \rightarrow \Sigma_s^*$ and $P_s^\Delta : \Delta^* \rightarrow \Delta_s^*$, respectively.

We can then distribute both products [Feng, 2007], i.e., $P_s^\Delta(\|_{i=1}^n R(w_i)) = \|_{i=1}^n P_s^\Delta R(w_i)$ and $P_s(\|_{i=1}^n w_i) = \|_{i=1}^n P_s(w_i)$. We have that $\|_{i=1}^n P_s(w_i) = \emptyset$ implies $\|_{i=1}^n P_s^\Delta R(w_i) = \emptyset$ since if different shared events, say $\sigma_1 \neq \sigma_2 \in \Sigma_s$ cause emptiness of $\|_{i=1}^n w_i$, then under Assumption A2, the corresponding different shared events $R(\sigma_1) \neq R(\sigma_2)$ cause emptiness of $\|_{i=1}^n P_s^\Delta R(w_i)$. Thus, $\|_{i=1}^n P_s(w_i) \neq \emptyset$, i.e., $\|_{i=1}^n w_i \neq \emptyset$ holds as well. Now, we need to show that there exists $w \in \|_{i=1}^n w_i$ such that $R(w) = s$. To see this, notice that for every $w \in \|_{j=1}^n w_j$ and $i = 1, \dots, n$, we have by Lemma 5 that $P_i^\Delta(s) = R_i(w_i) = R_i \circ P_i(w) = P_i^\Delta R(w)$. In simple words, the strings s and $R(w)$ have the same local projections. Since by Assumption A1 the relabeling $R(w)$ does not change the membership in the component alphabets (and therefore the synchronization status of all events in w), and the string $s \in \|_{i=1}^n P_i^\Delta(s)$, we can choose $w \in \|_{i=1}^n w_i$ such that the order of string in s is mimicked in w , yielding $R(w) = s$. More precisely, let $s = s_1 \dots s_n$ with $s_i \in \Sigma_{j_{k1}} \cap \dots \cap \Sigma_{j_{ki}}$, where $k_i \leq n$. Then, by Assumption A1, we have $R(s) = r_1 \dots r_n$, where $r_j = R(s_j) \in \Delta_{j_{k1}} \cap \dots \cap \Delta_{j_{ki}}$. $\square$

Lemma 9 If $\Sigma = \cup_{i=1}^n \Sigma_i$ is a distributed alphabet with Assumptions A1, A2, and A3, then for languages $L_i \subseteq \Sigma_i^*$ we have $\|_{i=1}^n M_i(L_i) \subseteq M(\|_{i=1}^n L_i)$.

Proof: It follows from $M = RP$ and the well known result that A3 implies $\|_{i=1}^n P_i(L_i) \subseteq P(\|_{i=1}^n L_i)$. $\square$

Note that the opposite inclusion does not hold, as it is known from supervisory control of multi-agent systems, where relabeling to a template alphabet from which all agent alphabets are instantiated is used.

In the following, based on the diagram shown in Fig. 2 and Definition 1, a mask consistency is proposed to guarantee that normality is preserved by masks.

Definition 10 A prefix-closed language $L \subseteq \Sigma^*$ is *mask-modified observation consistent* (MMOC) w.r.t. projections P_i , M , and M_i if, for every string $s \in L$ and $t' \in P_i(L)$ with $M_i(P_i(s)) = M_i(t')$, there exists a string $s' \in L$ such that $P_i(s') = t'$ and $M(s) = M(s')$. $\diamond$

The following proposition shows that $L(G)$ is MMOC under Assumptions A1–A3.

Proposition 11 Let $G = \|_{i=1}^n G_i$ be a modular DES over $\Sigma = \cup_{i=1}^n \Sigma_i$. Language $L(G)$ is MMOC w.r.t. P_i , M , and M_i if Assumptions A1–A3 hold.

Proof: Let $L = L(G)$ and $L_i = L(G_i)$. We need to show that for all $s \in L$ and $t' \in P_i(L)$ with $M_i(P_i(s)) = M_i(t')$, there is a string $s' \in L$ such that (i) $P_i(s') = t'$ and (ii) $M(s) = M(s')$. To see that such a sequence $s' \in L$ with conditions (i) and (ii) exists, we will define the following constraints on s' based on the given strings s and t' . Condition (i) defines $s' \in P_i^{-1}(t')$. Since $M = R \circ P$, condition (ii) means that $s' \in M^{-1}M(s) = P^{-1}R^{-1}RP(s)$, i.e., $Ps' \in R^{-1}RP(s)$. Finally, for $j \neq i$, we have from Lemma 7 that condition (ii) implies $M_j P_j(s') = M_j P_j(s)$. Since $M_j = R_j \circ P_{j,o}^j$, we obtain $P_j(s') \in M_j^{-1}(M_j P_j(s)) = P_{j,o}^{j-1} R_j^{-1} R_j P_{j,o}^j P_j(s)$. Otherwise stated, $s' \in R^{-1}RP(s) \|_{j \neq i} P_{j,o}^{j-1} R_j^{-1} R_j P_{j,o}^j P_j(s)$. We will show that this set is nonempty by showing that the mask of this set is nonempty.

It is known from [Feng, 2007, Lemma 4.1] that a language is nonempty iff its natural projection (to an arbitrary alphabet) is nonempty. This result partially extends to the mask, since the relabeling map only rewrites observable strings, and we only need the obvious implication that for $K \subseteq \Sigma_o^*$, $R(K) \neq \emptyset$ implies $K \neq \emptyset$. By Lemma 9, it suffices owing to Assumptions A1–A3 to show that $M(R^{-1}RP(s)) \|_{j \neq i} M_j(P_{j,o}^{j-1} R_j^{-1} R_j P_{j,o}^j P_j(s)) \neq \emptyset$. After substituting $M_j = R_j \circ P_{j,o}^j$ and knowing that both $P_{j,o}^j P_{j,o}^{j-1}$ and RR_j^{-1} are identity maps, $\|_{j \neq i} M_j(P_{j,o}^{j-1} R_j^{-1} R_j P_{j,o}^j P_j(s)) = \|_{j \neq i} R_j P_{j,o}^j P_j(s)$ holds. From $M_i(P_i(s)) = M_i(t')$, we have $R_i P_{i,o}^i P_i(s) =$

$$R_i P_{i,o}^i(t'). \text{ It holds } M_i(t') \parallel \|_{j \neq i} R_j P_{j,o}^j P_j(s) = \|_{i=1}^n R_i P_{i,o}^i P_i(s).$$

It is now easy to see that $M(s') = M(s) \in M(R^{-1}RP(s)) \parallel \|_{i=1}^n R_i P_{i,o}^i P_i(s)$, making this set nonempty. By Definition 10, $L(G)$ is MMOC. $\square$

Now we extend a result from Komenda and Masopust [2024], namely that mask-modified observation consistency guarantees that normality is preserved by natural projections, which is used in our main result, Theorem 13 on modularly computed supremal normal supervisors under observation masks.

Lemma 12 Given a DFA G with $L = L(G)$, if L is MMOC w.r.t. P_i , M , and M_i and $K \subseteq L$ is normal w.r.t. L and M , then $P_i(K)$ is normal w.r.t. $P_i(L)$ and M_i .

Proof: Since the inclusion $P_i(\bar{K}) \subseteq M_i^{-1}M_i(P_i(\bar{K})) \cap P_i(L)$ is trivial, it suffices to show $M_i^{-1}M_i(P_i(\bar{K})) \cap P_i(L) \subseteq P_i(\bar{K})$. Let $t' \in M_i^{-1}M_i(P_i(\bar{K})) \cap P_i(L)$. We need to show that $t' \in P_i(\bar{K})$. Then, there exists $s \in \bar{K}$ such that $M_i(P_i(s)) = M_i(t')$. By the MMOC assumption, there is a string $s' \in L$ such that $P_i(s') = t'$ and $M(s) = M(s')$, that is, $s' \in M^{-1}M(s) \cap L \subseteq M^{-1}M(\bar{K}) \cap L \subseteq \bar{K}$, where the last inclusion follows from the normality of K w.r.t. L and M . Hence, $t' = P_i(s') \in P_i(\bar{K})$ is true. $\square$

Theorem 13 Let $G = G_1 \| \dots \| G_n$ be a non-blocking modular DFA with $L = L(G)$ and $L_i = L(G_i)$. For any decomposable specification $K = \|_{i=1}^n K_i \subseteq L$,

$$\sup N(K, L, M) = \|_{i=1}^n \sup N(K_i, L_i, M_i) \text{ if}$$

- 1) $\sup N(K_i, L_i, M_i)$, $i = 1, \dots, n$ are non-conflicting,
- 2) Assumption A1 holds,
- 3) L is MMOC w.r.t. Q , M , and M_i , and
- 4) $P_i(L) = L_i$.

Proof: ($\subseteq$) This inclusion is equivalent to $P_i(\sup N(K, L, M)) \subseteq \sup N(K_i, L_i, M_i)$. Since $P_i(\sup N(K, L, M)) \subseteq P_i(K)$, it suffices to show that $P_i(\sup N(K, L, M))$ is normal w.r.t. L_i and M_i . This follows directly from Lemma 12.

($\supseteq$) We denote $\sup N(K, L, M)$ by $\sup N$ and $\sup N(K_i, P_i(L), M_i)$ by $\sup N_i$ to simplify the notation. It needs to be shown that $\|_{i=1}^n \sup N_i$ is normal w.r.t. L and M , from which the inclusion follows, because of $\|_{i=1}^n \sup N_i \subseteq \|_{i=1}^n K_i = K$. Let $s \in \overline{\|_{i=1}^n \sup N_i}$, and $s' \in L$ with $M(s) = M(s')$. By non-conflictingness assumption 1) we have $s \in \overline{\|_{i=1}^n \sup N_i}$, which is equivalent to $P_i(s) \in \overline{\sup N_i}$. From $s' \in L$ we have $P_i(s') \in L_i$. By $M(s) = M(s')$, we also have $P_i^\Delta M(s) = P_i^\Delta M(s')$. By Assumption A1 and Lemma 5, the commutativity in $M_i \circ P_i = P_i^\Delta \circ M$ holds true. Therefore, $M_i(P_i(s)) = M_i(P_i(s'))$, and from

normality of $\sup N_i$ w.r.t. L_i and M_i we obtain $P_i(s') \in \overline{\sup N_i}$, which means that $s' \in \overline{\|_{i=1}^n \sup N_i} = \overline{\|_{i=1}^n \sup N_i}$, where non-conflictingness has been used once more in the last equality. $\square$

Theorem 13 implies that under Assumption A1, MMOC, and non-conflictingness assumptions, supremal normal supervisors can be computed modularly. Based on Definitions 1 and 10, and Lemma 11, a more easily verifiable condition that guarantees the same conclusion is presented below.

Corollary 14 Let G be a modular DES $G = G_1 \| \dots \| G_n$ with $L = L(G)$, $L_i = L(G_i)$ and $L_m = L_m(G)$. Let $K = \|_{i=1}^n K_i \subseteq L_m$ be a decomposable specification. It holds $\sup N(K, L, M) = \|_{i=1}^n \sup N(K_i, L_i, M_i)$, if 1) $\sup N(K_i, L_i, M_i)$, $i = 1, \dots, n$ are non-conflicting, 2) Assumptions A1–A3 hold, and 3) $P_i(L) = L_i$.

Proof: It follows Theorem 13 and Lemma 11. $\square$

Corollary 14 indicates that under Assumptions A1–A3, we can compute the global supremal normal sublanguage $\sup N(K, L, M)$ w.r.t. plant L and mask M in a distributed, computationally cheaper way. This result can be easily extended by the well-known results in complete observations, and we can obtain the maximally permissive supervisor for the original plant based on distributed computation of the supervisors given by local supremal controllable and normal sublanguages.

Example 15 Consider a modular system consisting of two modules $G = G_1 \| G_2$ over the event set $\Sigma = \{a_1, b_1, a_2, b_2, c\}$, where each local component G_i over $\Sigma_i = \{a_i, b_i, c\}$ satisfies $L(G_i) = \{c^*, a_i, b_i\}$ for $i = 1, 2$. For simplicity, assume $\Sigma = \Sigma_o = \Sigma_c$, i.e., all events are observable and controllable. The observation mask M is defined by $M(a_1) = M(b_1) = \alpha$, $M(a_2) = M(b_2) = \beta$, and $M(c) = c$. Under this mask, the diagram of masks and local projections commutes by Lemma 5 since Assumption A1 is satisfied.

Given a global string $s = ca_1a_2$ and a local string $t' = ca_1 \in L(G_1)$ with $M_1P_1(s) = M_1(t') = c\alpha$, there exists a string $s' = ca_1b_2 \in L(G)$ such that $M(s) = M(s') = c\alpha\beta$ and $P_1(s') = t'$. Thus, $L(G)$ is MMOC. Consider a specification $K = K_1 \| K_2$, where $K_i = \{c^*, a_i\}$ for $i = 1, 2$. Then, K_i is not normal w.r.t. $L(G_i)$ and M_i for $i = 1, 2$ since there exists $cb_i \in L(G_i)$ with $M(cb_i) = M(ca_i)$ but $cb_i \notin K_i$. By Corollary 14, we have $\sup N(K, L, M) = \|_{i=1}^2 \sup N(K_i, L_i, M_i) = c^*$. $\blacksquare$

4 Synthesis of controllable and normal supervisors under masks for global specifications

In this section, we extend the previous results in two directions: (i) to simultaneously address controllability

and normality issues, (ii) to handle global specifications $K \subseteq L = \|_{i=1}^n L_i$, i.e., the case where K is not decomposable w.r.t. local alphabets. This constitutes the second problem to be addressed in this section.

Problem 16 Let $G = G_1 \| \dots \| G_n$ be a modular plant and $K \subseteq L(G) = \|_{i=1}^n L(G_i)$ the global specification that is not decomposable into local alphabets. Find sufficient conditions under which the global supremal normal and controllable sublanguage of K w.r.t. L and mask M equals the composition of local supremal normal and controllable sublanguages w.r.t. $L(G_i)$ and M_i . $\diamond$

In case the specification K is not decomposable into local alphabets, we can follow the approach in Komenda et al. [2012], K can be made conditionally decomposable, i.e., decomposable w.r.t. local alphabets union some coordinator events $\Sigma_i \cup \Sigma_k$, where Σ_k contains at least an event shared by two or more components.

To use modular control, we need in general to (conditionally) decompose K w.r.t. local alphabets Σ_i , $i = 1, \dots, n$. This can be done by extending the shared event set $\Sigma_s = \cup_{j \neq i} \Sigma_i \cap \Sigma_j$ to get so-called coordinator alphabet $\Sigma_k \supseteq \Sigma_s$ such that K is decomposable w.r.t. “coordinated” local alphabets $\Sigma_{i+k} = \Sigma_i \cup \Sigma_k$, $i = 1, \dots, n$. K is then called conditionally decomposable w.r.t. local alphabets Σ_i , $i = 1, \dots, n$ and coordinator alphabet Σ_k , i.e., $K = \|_{i=1}^n P_{i+k}(K)$, where $P_{i+k} : \Sigma^* \rightarrow \Sigma_{i+k}^*$. Let $\Sigma_{i+k,uc} = \Sigma_{i+k} \cap \Sigma_{uc}$ be the set of coordinated local uncontrollable events.

The corresponding coordinator language (plant) is then given by $L_k = \|_{i=1}^n P_k(L_i) = P_k(L)$, i.e., projection of the plant to the coordinator alphabet computed in a distributed way, and the coordinated local plants are given by $L_i \| L_k$ for $i = 1, \dots, n$. We then simply enforce normality of $P_{i+k}(K)$ w.r.t. $L_i \| L_k$ and restriction of the mask M on $(\Sigma_i \cup \Sigma_k)^*$, denoted by M_{i+k} . Since we are using supervisory control, we have to compute the sublanguage of $P_{i+k}(K)$ that is controllable and normal. In case all controllable events are observable, supremal controllable and normal sublanguages, denoted $\sup CN(P_{i+k}(K), L_i \| L_k, \Sigma_{i+k,uc}, M_{i+k})$, represent maximally permissive local enforcement of both normality and observability [Komenda et al., 2012]. Naturally, the same conditions as in Theorem 13 and Corollary 14 guarantee that these local maximally permissive supervisors enforcing normality running in parallel equal to the maximally permissive global (monolithic) supervisor enforcing normality, i.e. $\sup CN(K, L, \Sigma_{uc}, M) = \|_{i=1}^n \sup CN(P_{i+k}(K), L_i \| L_k, \Sigma_{i+k,uc}, M_{i+k})$. Namely, we still need either the MMOC condition or the sufficient conditions, i.e., Assumptions A1–A3.

To enforce mask normality aspects of maximal permissiveness in conjunction with the controllability aspect of maximal permissiveness, we use the observer [Wong and Wonham, 1996] and local control consistency [Schmidt

and Breindl, 2011] properties. The projection $P : \Sigma^* \rightarrow \Sigma_o^*$ has the observer property for a non-blocking plant G over Σ if for all strings $s \in \overline{L(G)}$ such that $P(s)a \in P(L(G))$ for some $a \in \Sigma_o$, there is a string $u \in \Sigma^*$ such that $su \in L(G)$ and $P(u) = a$. The projection $P : \Sigma^* \rightarrow \Sigma_o^*$ is local control consistent for a string $s \in L(G)$ if for every $e \in \Sigma_o \cap \Sigma_{uc}$ with $P(s)e \in P(L(G))$, either there is no string $u \in (\Sigma \setminus \Sigma_o)^*$ such that $sue \in L(G)$, or there exists a string $u \in (\Sigma_{uc} \setminus \Sigma_o)^*$ satisfying $sue \in L(G)$. The projection is local control consistent for a language L if it is true for all strings $s \in L$.

Let G over Σ be a non-blocking plant, K over $\Sigma_o \subseteq \Sigma$ be a specification, and $P : \Sigma^* \rightarrow \Sigma_o^*$ be the corresponding projection. It is known from Schmidt and Breindl [2011] that given $\text{supC}(K, L(G), \Sigma_{uc}, P)$, the supremal controllable sublanguage of $K \parallel L_m(G)$ w.r.t. $L(G)$ and Σ_{uc} , and the projection P , satisfies the observer property and local control consistent for $L(G)$, then the language $P(\text{supC}(K, L(G), \Sigma_{uc}, P))$ is controllable w.r.t. $P(L(G))$ and $\Sigma_{uc} \cap \Sigma_o$. We denote by $\text{supCN}(K, L, \Sigma_{uc}, M)$ the supremal controllable and normal sublanguage of the specification K w.r.t. L, Σ_{uc} and M .

Theorem 17 For an integer $n \geq 2$, consider a modular DFA G with $L(G) = L = \parallel_{i=1}^n L_i$ over $\Sigma = \cup_{i=1}^n \Sigma_i$, and a global specification $K \subseteq L$, which is not necessarily decomposable. Suppose that there exists an alphabet Σ_k containing all shared events such that Σ_k makes the specification K conditionally decomposable w.r.t. alphabets $\Sigma_i, i = 1, \dots, n$ and Σ_k . It holds

$$\text{supCN}(K, L, \Sigma_{uc}, M) = \parallel_{i=1}^n \text{supCN}(P_{i+k}(K), L_i \parallel L_k, \Sigma_{i+k,uc}, M_{i+k}) \quad (1)$$

if for all $i = 1, 2, \dots, n$,

- 1) languages $\text{supCN}(P_{i+k}(K), L_i \parallel L_k, \Sigma_{i+k,uc}, M_{i+k})$ are non-conflicting,
- 2) the projections $P_{i+k} : \Sigma^* \rightarrow \Sigma_{i+k}^*$ have the observer and local control consistent properties for L , and
- 3) Assumptions A1–A3 hold.

Proof: The preservation of normality follows from Corollary 14. To show the preservation of controllability, we consider two directions.

($\subseteq$) Since $\text{supCN}(K, L, \Sigma_{uc}, M) \subseteq K$ is controllable w.r.t. L and Σ_{uc} , by [Schmidt and Breindl, 2011, Lemma 4.1], $P_{i+k}(\text{supCN}(K, L, \Sigma_{uc}, M)) \subseteq P_{i+k}(K)$ is controllable w.r.t. $P_{i+k}(L) = L_i \parallel L_k$ and $\Sigma_{i+k,uc} = (\Sigma_i \cup \Sigma_k) \cap \Sigma_{uc}$. Together with the similar normality argument from Theorem 13 this implies $P_{i+k}(\text{supCN}(K, L, \Sigma_{uc}, M)) \subseteq \text{supCN}(P_{i+k}(K), L_i \parallel L_k, \Sigma_{i+k,uc}, M_{i+k})$, i.e., for all $i = 1, \dots, n$ $\text{supCN}(K, L, \Sigma_{uc}, M) \subseteq P_{i+k}^{-1}(\text{supCN}(P_{i+k}(K), L_i \parallel L_k, \Sigma_{i+k,uc}, M_{i+k}))$ holds, from which the inclusion follows.

($\supseteq$) This inclusion can be done similarly as in Schmidt

and Breindl [2011] by using the fact from Cassandras and Lafortune [2021] that the synchronous composition of the controllable languages is controllable if non-conflictingness is assumed, in the same way as it holds for normality, cf. Theorem 13. $\square$

It is shown in Kumar et al. [1991] that the complexity of the computation of the supremal controllable and normal sublanguage of a specification $K \subseteq L$ w.r.t. L and P is $O(2^{mr})$ for prefix-closed languages, where m and r are the numbers of states in the minimal generator representations of L and K , respectively. Given a modular DES $G = \parallel_{i=1}^n G_i$ and the global specification K , denote the number of states of minimal generators for $L(G_1), L(G_2), \dots, L(G_n), L(G_k)$ by $m_1, m_2, \dots, m_n, m_k$, respectively, where $n \in \mathbb{N}$ and G_k is the coordinator automaton. In this way, due to $K = \parallel_{i=1}^n P_{i+k}(K)$ and observer property, the number of states of minimal generators for $P_{i+k}(K)$ and $P_k(K)$ should be bounded by r , where $i \in 1, 2, \dots, n$. Then, in the worst case, the computational complexity of $\parallel_{i=1}^n \text{supCN}(P_{i+k}(K), L_i \parallel L_k, \Sigma_{i+k,uc}, M_{i+k})$ results in $O(\sum_{i=1}^n 2^{m_i m_k r})$, which is significantly lower than the monolithic complexity of $\text{supCN}(K, L, \Sigma_{uc}, M)$, i.e., $O(2^r \prod_{i=1}^n m_i)$ provided m_k is significantly smaller than m , i.e., the size of the coordinator is much smaller than that of the global plant, which is usually in case.

In the following, we present a motivating example to illustrate the proposed approaches in detail. Before constructing the supremal controllable and normal sublanguages $\text{supCN}(K, L, \Sigma_{uc}, M)$, we first introduce several necessary notions and notations.

Given a DFA $G = (Q, \Sigma, \delta, q_0)$, its masked automaton is a non-deterministic finite automaton $M(G) = (Q, \Delta \cup \{\varepsilon\}, \delta_m, q_0)$, where Q and q_0 , as before, are the state set and initial state, respectively, $\Delta \cup \{\varepsilon\}$ is the set of events and $\delta_m \subseteq Q \times \Delta \cup \{\varepsilon\} \times Q$ is a transition relation such that $(q, \alpha, q') \in \delta_m$ iff there exists $e \in \Sigma$ with $M(e) = \alpha$ such that $q' = \delta(q, e)$. Then, given a DFA G and a mask M , a mask observer, denoted by $Obs_m(G)$, is constructed by $Obs_m(G) = Det(M(G))$, where $Det(\cdot)$ is special determinisation operation known as ε -removal [Hopcroft et al., 2006]. An automaton G is a *state-partition automaton* (SPA) w.r.t. M if any two states of its (mask) observer do not have a nontrivial overlap.

Example 18 In modern manufacturing systems, multiple robots often share a few high-precision workstations, and their behaviors can be modeled as discrete event systems. Each robot executes four operations: requesting a workstation, performing its task, withdrawing before execution, and releasing the resource. Since several robots may request access simultaneously, arbitrary interleavings can lead to unsafe or inconsistent behaviors. Supervisory control is therefore needed to prevent incor-

rect sequences and ensure safe coordination.

Consider a manufacturing setting with three robots sharing a high-precision workstation. The overall behavior of the system is modeled by the plant $G = G_1 \| G_2 \| G_3$ over the set of events $\Sigma = \{r_1, r_2, r_3, a_1, a_2, a_3, w_1, w_2, w_3, e_1, e_2, e_3\}$, where local component G_i is depicted in Fig. 3(a) for $i = 1, 2, 3$. For simplicity, let $\Sigma = \Sigma_o$ and $\Sigma_c = \{a_1, a_2, a_3, e_1, e_2, e_3\}$, i.e., all events are observable and the supervisor can restrict the progress of each robot. The specification K portrayed in Fig. 3(b), captures the correct coordination requirement, i.e., a robot must complete its current access-exit cycle before any other robot is allowed to enter the access phase. This prevents unsafe overlaps in workstation usage and ensures mutually exclusive access among the three robots.

To capture the realistic situation where the supervisor cannot distinguish which robot is currently accessing the workstation, we introduce an observation mask M . Specifically, the three access events are treated as indistinguishable, and therefore their labels are merged under the mask, i.e., $M(a_1) = M(a_2) = M(a_3) = a$ and $M(b) = b$ for $b \in \Sigma \setminus \{a_1, a_2, a_3\}$.

It is straightforward to verify that the global specification K is not decomposable w.r.t. Σ_i . To ensure conditional decomposability, the coordinator must include the shared access events, that is, $\Sigma_k = \{a_1, a_2, a_3\}$. Using this coordinator alphabet, the coordinator automaton $G_k = \parallel_{i=1}^3 P_k(G_i)$ is constructed as shown in Fig. 4(a). By the definition of $P_{i+k} : \Sigma^* \rightarrow \Sigma_{i+k}^*$, the projected specifications $P_{i+k}(K)$ for $i = 1, 2, 3$ are obtained and illustrated in Fig. 4(b). Consequently, K is conditionally decomposable w.r.t. Σ_i and Σ_k since $K = \parallel_{i=1}^3 P_{i+k}(K)$ holds.

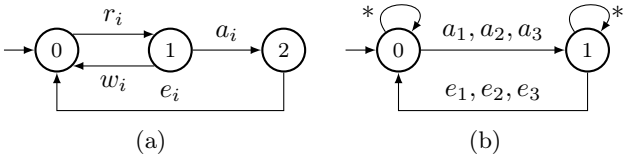


Fig. 3. (a) DFA G_i for $i = 1, 2, 3$ and (b) the global specification K , where $*$ represents events $r_1, r_2, r_3, w_1, w_2, w_3$.

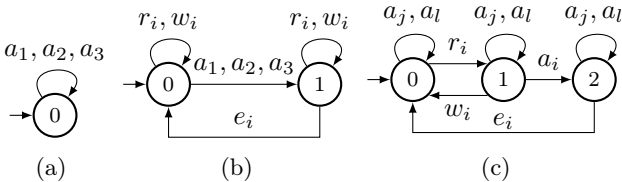


Fig. 4. (a) The coordinator G_k , (b) $P_{i+k}(K)$, and (c) $G_i \| G_k$, where $i, j, l \in \{1, 2, 3\}$ and $i \neq j \neq l \neq i$.

The composed generators $G_i \| G_k$ for $i = 1, 2, 3$ are portrayed in Fig. 4(c). Now, we show how to compute the supremal controllable and normal sublanguage w.r.t. $L(G_i)$, $\Sigma_{i,uc}$ and M , i.e., $\text{supCN}(P_{i+k}(K), L_i \| L_k, \Sigma_{i+k,uc}, M_{i+k})$. For any component G_i , we first construct an unfolded automaton H_i with $L(H_i) = L(G_i \| G_k)$ and its mask observer $\text{Obs}_m(H_i)$ as shown in Figs. 5 and 6, respectively. Note that H_i is not an SPA w.r.t. M , since certain observer states overlap, e.g., $\{2, 4\} \cap \{4\} \neq \emptyset$ and $\{5, 7\} \cap \{7\} \cap \{7, 8\} \neq \emptyset$. Nevertheless, the supremal mask-normal language can be computed using this (non-SPA) representation since a weaker property holds, which is still sufficient for computing the supremal mask-normal language. Specifically, this property requires that all observer states violating the SPA condition share exactly the same future observable behavior. This is obvious in this example, since these observer states lead to the same next state for all possible (defined) transitions in $\text{Obs}_m(H_i)$. Specifically, after firing event a , states $\{2, 4\}$ and $\{4\}$ both reach $\{5, 7\}$. State $\{3\}$ is reached by firing w_i from both states $\{2, 4\}$ and $\{4\}$. Similarly, by firing a , states $\{5, 7\}$, $\{7\}$, and $\{7, 8\}$ all reach $\{6\}$. States $\{5, 7\}$ and $\{7, 8\}$ subsequently reach $\{6\}$ by firing e_i and w_i . Therefore, there will be no conflicting decisions based on different states of the observer.

From Lemma 11, $L(G)$ is MMOC. since $M(a_1) = M(a_2) = M(a_3)$, i.e., a_1, a_2 , and a_3 cannot be distinguished under mask M . Given the projected specifications $P_{i+k}(K)$ depicted in Fig. 4(b), one needs to disable a at observer states $\{2, 4\}$, $\{3\}$, and $\{4\}$. The local supremal controllable and normal sublanguage $\text{supCN}(P_{i+k}(K), L_i \| L_k, \Sigma_{i+k,uc}, M_{i+k})$ is shown in Fig. 7(b). It is obvious that $\text{supCN}(P_{i+k}(K), L_i \| L_k, \Sigma_{i+k,uc}, M_{i+k})$, for $i = 1, 2, 3$, are non-conflicting. Further, projection P_{i+k} has the observer and local control consistent properties for L . Finally, from Theorem 17, the supremal controllable and normal sublanguage of the specification K w.r.t. L , Σ_{uc} , and M , equals to $\parallel_{i=1}^3 \text{supCN}(P_{i+k}(K), L_i \| L_k, \Sigma_{i+k,uc}, M_{i+k})$. ■

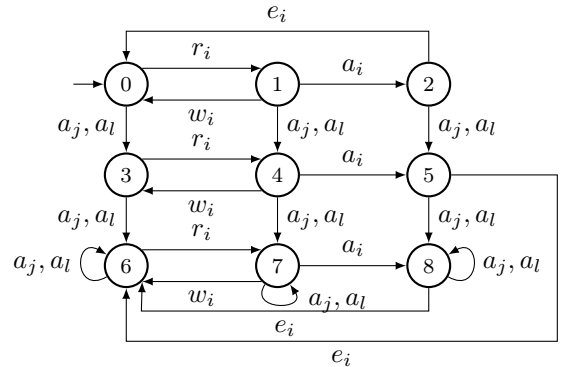


Fig. 5. The unfolded automaton H_i with $L(H_i) = L(G_i \| G_k)$, where $i, j, l \in \{1, 2, 3\}$ and $i \neq j \neq l \neq i$.

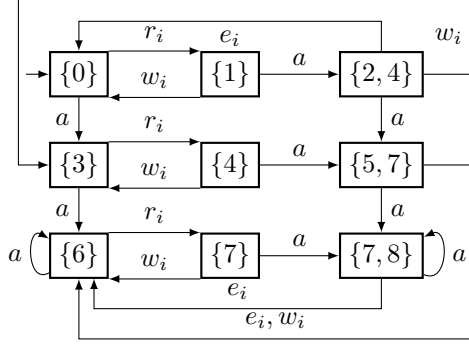


Fig. 6. Mask observer $Obs_m(H_i)$ for $i \in \{1, 2, 3\}$.

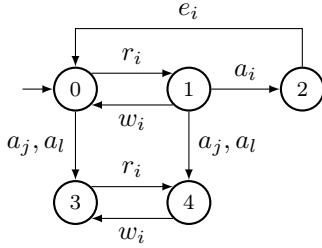


Fig. 7. The supremal controllable and normal sublanguage $\sup CN(P_{i+k}(K), L_i || L_k, \Sigma_{i+k, uc}, M_{i+k})$, where $i, j, l \in \{1, 2, 3\}$ and $i \neq j \neq l \neq i$.

5 Conclusion

This article investigates modular control of discrete event systems under partial observations within a mask-based observation framework. A necessary and sufficient condition is established to guarantee commutativity between observation masks and local projections. For decomposable specifications, we show that if Assumptions A1–A3 are satisfied, the global supremal normal supervisor with respect to the plant and mask coincides with the synchronous composition of local, non-conflicting, supremal normal supervisors. Furthermore, this result is extended to global (non-decomposable) specifications, as well as to the joint treatment of controllability and normality, by means of a well-known coordinator alphabet approach.

Finally, it is worth noting that the mask-based observation framework enables seamless integration of the proposed results with existing modular methodologies, potentially yielding additional computational benefits. A natural direction for future research is the systematic unification of these techniques, with particular emphasis on modular and multi-agent control architectures operating under masks, including scenarios involving non-deterministic observations that may arise in networked DESs or under sensor attacks.

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